

WEEK 2 – TESTING DOCUMENTATION

Educational Microcontroller Simulator – MicroOS / STC89C52

1. Testing Objective

The objective of testing is to verify that the implemented instruction set executes correctly, that CPU registers and flags change as expected, that program-counter control instructions behave correctly, and that the complete demonstration program executes and terminates successfully.

2. Testing Method

The MicroOS simulator test suite uses small Java test programs. Each test creates a simulator instance, loads a short instruction sequence, executes it using Step where required, and checks the resulting CPU state. A separate demonstration-program test loads the complete instruction sequence and verifies the final register values and program termination status.

3. Individual Instruction Test Cases and Results

Test ID	Instruction	Test Program / Input	Expected Result	Actual Result	Status
TC01	MOV A,#05	MOV A,#05 ONJI BYTE	A = 05H	A = 05H	PASS
TC02	MOV R0,#03	MOV R0,#03	R0 = 03H	R0 = 03H	PASS
TC03	XCH A,R0	MOV A,#05 MOV R0,#03 XCH A,R0	A = 03H; R0 = 05H	A = 03H; R0 = 05H	PASS
TC04	ADD A,#03	MOV A,#05 ADD A,#03	A = 08H	A = 08H	PASS
TC05	SUBB A,#02	MOV A,#08 SUBB A,#02	A = 06H	A = 06H	PASS
TC06	ANL A,#03	MOV A,#0F ANL A,#03	A = 03H	A = 03H	PASS
TC07	ORL A,#01	MOV A,#06 ORL A,#01	A = 07H	A = 07H	PASS
TC08	INC A	MOV A,#05 INC A	A = 06H	A = 06H	PASS
TC09	DEC A	MOV A,#05 DEC A	A = 04H	A = 04H	PASS
TC10	CLR A	MOV A,#05 CLR A	A = 00H; Zero Flag = true	A = 00H; Zero Flag = true	PASS
TC11	SJMP 1	MOV A,#05 SJMP 1 CLR A END	PC = 03H; CLR A skipped	PC = 03H	PASS

4. Complete Demonstration Program Test

Demonstration Program:

MOV A, #05
MOV R0, #03
XCH A, R0
ADD A, #03
INC A
DEC A
ANL A, #0F
ORL A, #01
SJMP 1
CLR A
END

Test ID	Expected Result	Actual Result	Status
TC12	A = 07H; R0 = 05H; program terminated	A = 07H; R0 = 05H; status = Program terminated	PASS

5. Overall Testing Result

The repository contains 11 individual instruction test cases covering MOV, register MOV, XCH, ADD, SUBB, ANL, ORL, INC, DEC, CLR and SJMP. The complete demonstration-program test checks the final accumulator, R0 register and program termination status.

Result represented by the repository test definitions: 11 / 11 individual instruction cases PASS; complete demonstration program PASS.

6. Testing Summary

Testing Item	Result
Individual instruction tests	11 / 11 PASS
Complete demonstration program	PASS
Overall testing result	PASS